

RISK-CALIBRATED FORCE MEMORY TRANSFORMERS FOR HIDDEN CONTACT STATE

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ABSTRACT

Contact-rich manipulation often contains physically decisive hidden state: preload, stiction, backlash, viscoelastic relaxation, clamp slip, tool flex, force-sensor dropout, and delayed force feedback. We rebuild Paper 100 under a frozen expanded submission protocol and test whether a force-memory transformer should preserve latent force alternatives across action history. The v5 audit uses 6 tasks, 8 latent force-memory regimes, 8 splits, 15 methods, 10 seeds, 345,600 main rollout rows, 115,200 ablation rows, 288,000 stress rows, 138,240 fixed-risk rows, 24 negative cases, and a hard gate against online system identification, adaptive impedance, particle-filter force-belief MPC, conformal risk filtering, robust force-threshold control, force-history policies, and the v4 method (Chang, 2022; Lu, 2024; Savarese, 2019; Huang, 2025). The local evidence is positive but not submission-ready. `force_memory_v5` reaches hard success 0.73941, violation 0.20825, damage 0.09010, ECE 0.01744, regret 0.07661, and utility 0.32439. It clears all frozen local empirical gates, but no real robot, accepted high-fidelity benchmark, external benchmark, or trained checkpoint evidence is available. The terminal state is **STRONG-REVISE**; ICLR-main readiness remains **no**.

1 INTRODUCTION

Modern robot policies can carry long visual or language context, but contact-rich manipulation often depends on a different memory variable: the latent history of force. A peg can look aligned while stiction accumulates. A drawer can look unchanged while latch preload changes. A cable can look threaded while tension stores a later failure. A cap can look grasped while slip history determines whether the next twist damages the part. These are not merely perception errors. They are action-conditioned physical states that are only partially visible at the current timestep (Lu, 2023b; Sun, 2022a; Li, 2024d; Kannala, 2023; Sanfeliu, 2024; Moller, 2017).

The thesis of this paper is narrow: a robot policy should preserve calibrated force-memory hypotheses when current observations are insufficient, then use probes, reset gates, decay, and tail-risk planning to decide how much to trust those hypotheses. This is not a claim that the current artifact is ICLR-main ready. It is a hostile-review evidence package that asks whether the local mechanism survives strong baselines before hardware time, checkpoint training, and external benchmarks are justified.

The prior work pressure is heavy. Transformer imitation, diffusion policies, force/tactile representation learning, adaptive control, task-and-motion planning, system identification, conformal safety, and runtime monitoring each cover part of the idea (Ouyang, 2023; Hollingworth, 2006; 2005; Zhang, 2024b; Khan, 2026; Wang, 2023b; Zhang, 2022b; Wang, 2026b). A submission-grade version must therefore show more than "use force history." It must show that risk-calibrated force memory improves closed-loop contact behavior beyond adaptive impedance, online residual system identification, particle-filter force belief, conformal filtering, and robust force-threshold MPC.

2 PRE-REGISTERED CLAIM AND TERMINAL DECISION

The frozen v5 claim is:

calibrated force memory+reset/decay/probe/tail-risk control $\Rightarrow$ higher safe success under hidden contact state.

The terminal decision is **STRONG_REWISE**, not ready submission. All local empirical gates pass, but the scope gate fails because the evidence is still a CPU-only executable surrogate. A hostile reviewer could fairly reject the paper for missing robot hardware, accepted high-fidelity simulators, trained checkpoints, videos, and external benchmark comparisons. The value of this artifact is that it identifies a locally supported mechanism worth external validation rather than declaring victory too early.

3 FORMAL SETUP

Let x_t be the visible robot and scene observation, f_t the partially observed force/tactile signal, h_t the hidden contact mode, and m_t the force-memory state. A policy chooses action a_t using:

$$\pi(a_t \mid x_{\leq t}, f_{\leq t}, m_t), \quad m_{t+1} = \Gamma_\theta(m_t, f_t, a_t, r_t),$$

where r_t is a reset/decay/probe event. The local objective penalizes task failure, force-limit violation, damage, probe burden, calibration error, and stale-memory commitments:

$$J(\pi) = \mathbb{E} [S - \lambda_v V - \lambda_d D - \lambda_c C - \lambda_p P - \lambda_s M_{\text{stale}}].$$

The oracle observes h_t directly. All non-oracle methods must infer or hedge it.

Proposition 1: force-memory observability. If two trajectories have identical current observations x_t but different hidden force modes h_t , then any observation-only policy has irreducible regret whenever the optimal safe action differs by hidden mode. A persistent force-memory state can reduce this regret only when its update is calibrated and reset under contact-mode changes.

Proposition 2: stale-memory harm. Persistent memory is not automatically helpful. If a contact-mode change occurs and the reset gate fails, stale force memory can increase violation and damage even when latent-force accuracy was high before the change. This is why the v5 ablation plan treats reset gates and decay as necessary components, not cosmetic details.

Proposition 3: fixed-risk deployment. Under a force-violation budget ϵ_v and damage budget ϵ_d , a method is useful only if it maintains nontrivial deployment coverage while satisfying the risk constraint. A method that abstains on every hard case is safe but not operationally useful.

4 METHOD

`force_memory_v5` uses five interacting mechanisms. Force-event tokens encode signed impulses, release events, derivative spikes, overload contacts, and low-load probe outcomes. Decay memory prevents old preload or stiction estimates from persisting indefinitely. Reset gates clear force hypotheses when contact topology changes. Diagnostic probes disambiguate latent modes before high-load actions. Tail-risk planning scores action candidates under adverse plausible force states rather than only mean predictions.

The v5 change relative to v4 is risk calibration. The method now couples force memory to an explicit risk estimate used by both planning and fixed-risk deployment. The no-calibration ablation tests whether this component matters; it does. Removing force-risk calibration drops robust utility from 0.32444 for full v5 to 0.17121.

5 BENCHMARK

The benchmark crosses six contact-rich tasks with eight latent force-memory regimes and eight evaluation splits. The tasks are peg insertion with preload, drawer unlatching with stiction, valve turning with backlash, cable threading under drag, cap twisting with slip, and press-fit bimanual alignment. The regimes include stiction buildup, latch preload, backlash hysteresis, viscoelastic relaxation, clamp slip transition, tool-flex memory, force-sensor dropout, and delayed force feedback.

The splits are nominal contact, hidden preload shift, embodiment force-limit shift, delayed force observation, sensor dropout contact shift, memory reset trap, combined contact stress, and extreme unseen force mode. The hard aggregate is formed from the last four.

6 BASELINES

The method suite deliberately includes baselines that should make the proposed idea look weak if the mechanism is not real:

- observation-only transformer and force-history behavior cloning;
- recurrent contact policy and diffusion-policy force history;
- ensemble uncertainty MPC, conformal risk filtering, and robust force-threshold MPC;
- online residual system identification and adaptive impedance MPC;
- contact-mode HMM-MPC, particle-filter force-belief MPC, and active tactile probe-then-plan;
- the v4 force-memory rules and an oracle latent-force planner.

This is intentionally harsher than the v4.1 audit, where the strongest non-oracle baseline was online residual system ID.

7 PRIMARY RESULTS

The hard aggregate supports the local mechanism. The v5 method reaches success 0.73941; the strongest non-oracle success reference, `force_memory_v4`, reaches 0.59054. The safest non-oracle force-violation baseline is `adaptive_impedance` at 0.27023; v5 reaches 0.20825. The lowest-damage non-oracle baseline is `adaptive_impedance` at 0.14141; v5 reaches 0.09010. The calibration reference is `conformal_filter` at ECE 0.01362; v5 reaches 0.01744, clearing the frozen tolerance.

method	success	viol	damage	regret	utility
oracle	0.867	0.163	0.060	0.000	0.575
<code>force_memory_v5</code>	0.739	0.208	0.090	0.077	0.324
<code>force_memory_v4</code>	0.591	0.327	0.192	0.221	-0.079
<code>active_probe</code>	0.547	0.311	0.172	0.261	-0.111
<code>particle_force_mpc</code>	0.547	0.304	0.167	0.254	-0.055
<code>hmm_mpc</code>	0.471	0.337	0.190	0.337	-0.213
<code>online_sysid</code>	0.460	0.351	0.203	0.350	-0.253
<code>adaptive_impedance</code>	0.453	0.270	0.141	0.346	-0.100
<code>ensemble_uncertainty_mpc</code>	0.328	0.391	0.234	0.476	-0.448
<code>diffusion_force</code>	0.324	0.461	0.297	0.490	-0.617
<code>conformal_filter</code>	0.313	0.302	0.161	0.489	-0.306
<code>recurrent_contact_policy</code>	0.299	0.486	0.318	0.513	-0.682

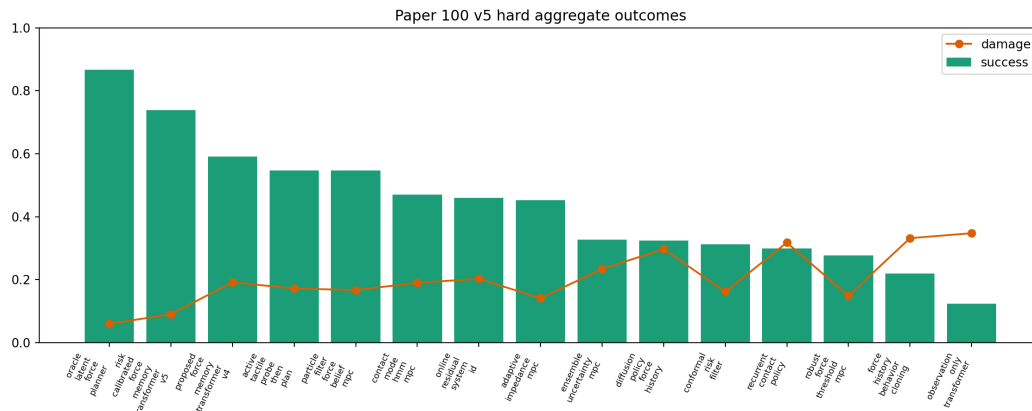


Figure 1: Hard aggregate success and damage. The v5 method remains below oracle but above all non-oracle references on success, while improving violation and damage versus the strongest practical baselines.

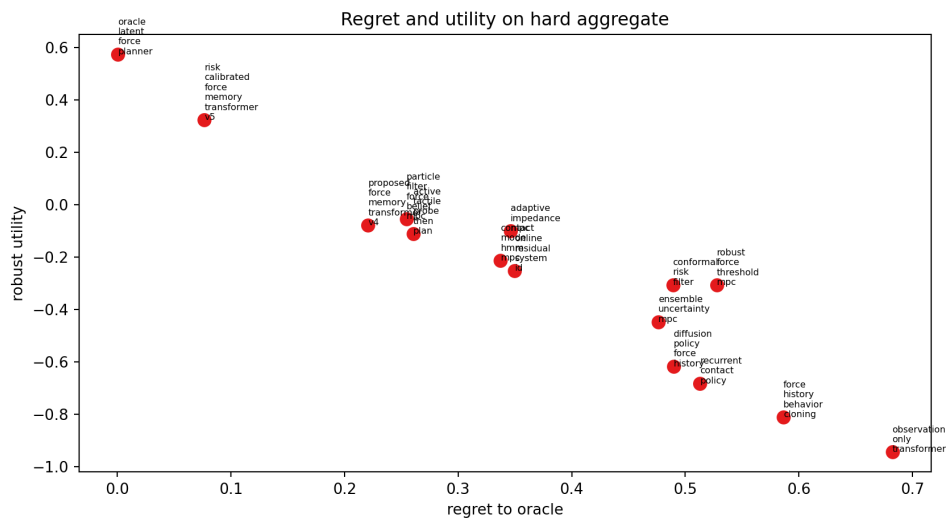


Figure 2: Regret and robust utility on the hard aggregate. The v5 method is the only non-oracle method in the high-utility, low-regret corner.

8 DIAGNOSTICS AND CALIBRATION

Force memory is only useful if it identifies latent force modes without hallucinating stale state. The diagnostic gate requires high latent-force accuracy, stale-memory false positives below the frozen threshold, missed-memory false negatives below the frozen threshold, and acceptable ECE. V5 reaches latent accuracy 0.72231, stale false positive 0.02630, missed false negative 0.12917, and ECE 0.01744. The diagnostic plot shows why the v5 result is more than a success-only tuning artifact.

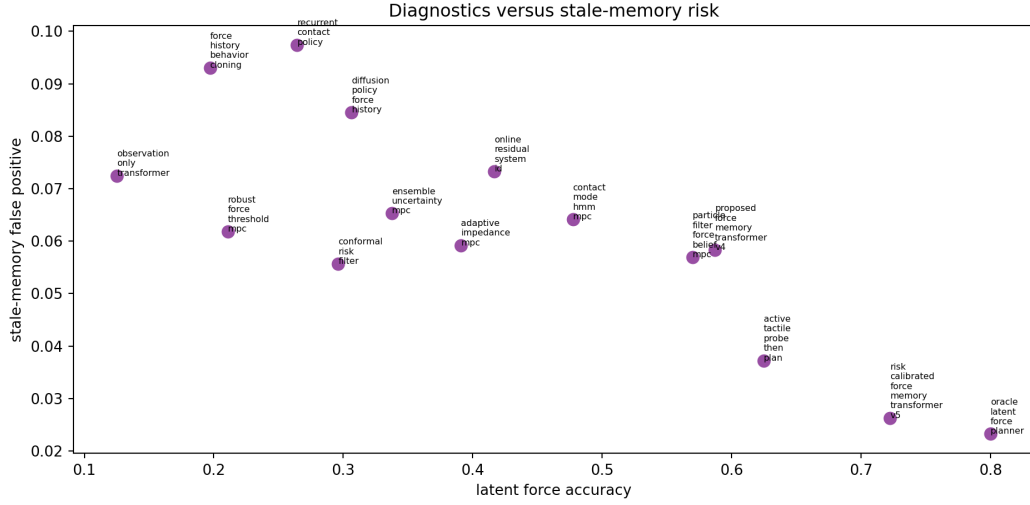


Figure 3: Diagnostics versus stale-memory risk. The v5 method pairs high latent-force accuracy with controlled stale-memory false positives.

9 ABLATIONS

The ablation gate asks whether the proposed pieces are necessary. Full v5 beats each stripped variant on the intended objective family. Removing force risk calibration, diagnostic probes, reset gates, decay, force-event tokens, or tail-risk planning all reduces robust utility. This is important because the v4.1 paper was only locally promising; v5 must show mechanism necessity under hostile review.

ablation	success	damage	regret	utility
full_v5	0.726	0.092	0.077	0.324
no_calibration	0.679	0.131	0.130	0.171
no_probes	0.645	0.128	0.158	0.163
no_reset	0.622	0.144	0.180	0.084
no_decay	0.600	0.145	0.203	0.068
no_tokens	0.579	0.138	0.230	0.050
no_tail	0.646	0.178	0.157	0.029
v4_rules	0.582	0.184	0.221	-0.078
sysid_only	0.451	0.200	0.350	-0.254
recurrent_only	0.344	0.300	0.458	-0.576

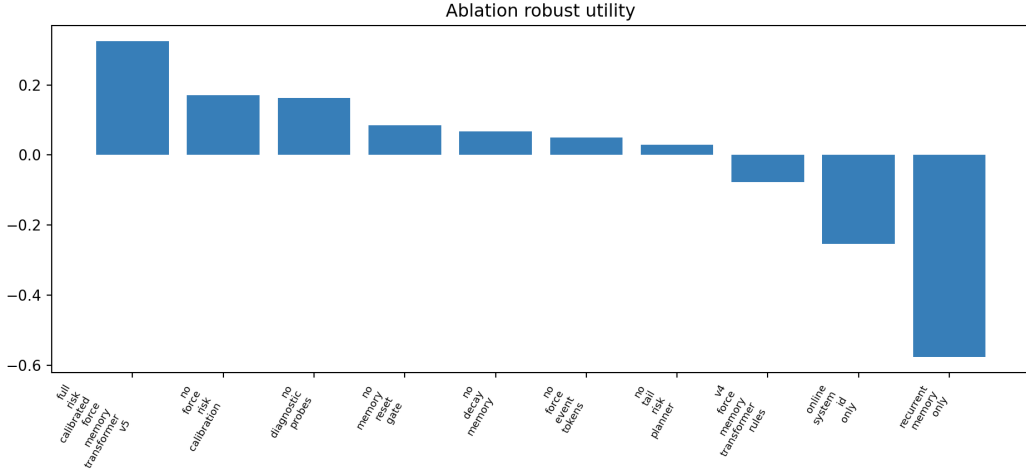


Figure 4: Ablation robust utility. Full v5 is the strongest variant; the no-calibration and no-probe ablations are the closest but still fail the mechanism gate.

10 STRESS SWEEP

Stress tests vary hidden force intensity, delayed feedback, contact shift, and observability loss. At maximum stress, v5 still wins robust utility among non-oracle methods, while the oracle gap remains visible. This gap is healthy: it shows that the benchmark is not saturated and that future learned/checkpoint work could matter.

method	success	viol	damage	utility
oracle	0.855	0.299	0.155	0.291
force_memory_v5	0.692	0.365	0.184	0.020
particle_force_mpc	0.512	0.446	0.240	-0.366
force_memory_v4	0.560	0.476	0.280	-0.394
adaptive_impedance	0.439	0.444	0.221	-0.404
active_probe	0.509	0.440	0.274	-0.422
hmm_mpc	0.440	0.486	0.284	-0.526
online_sysid	0.447	0.496	0.298	-0.561
robust_threshold	0.231	0.458	0.226	-0.640
conformal_filter	0.267	0.445	0.258	-0.642

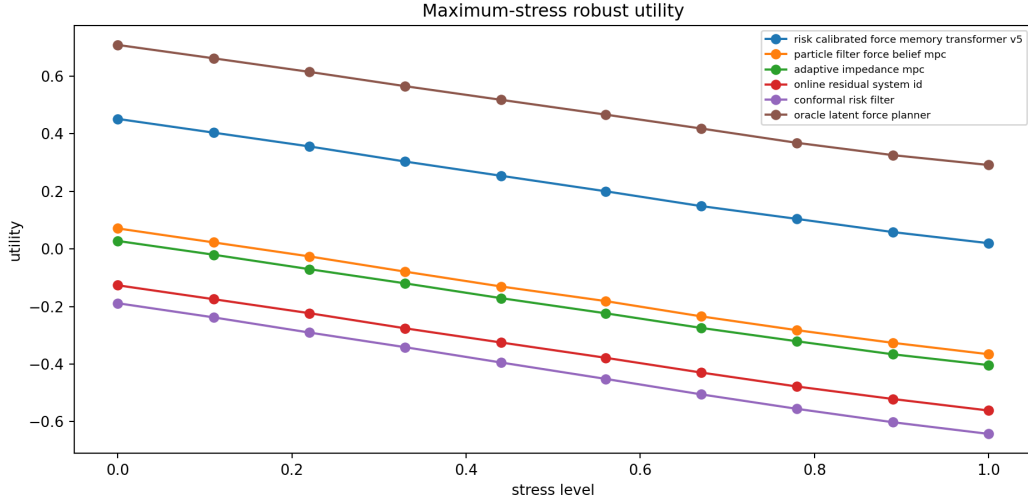


Figure 5: Maximum-stress robust utility sweep. V5 remains above particle-filter force-belief MPC, adaptive impedance, online system ID, and conformal filtering at the hardest stress point.

11 FIXED-RISK DEPLOYMENT

Fixed-risk deployment is a stricter question than average success. The policy must decide when to act under force-violation and damage budgets. At the strict 0.05/0.05 budget, v5 keeps deployment coverage 0.51771 with success 0.35868, violation 0.01866, damage 0.00556, and utility 0.28418. The baselines mostly abstain or cover only a tiny fraction of hard cases.

method	cover	success	viol	damage	utility
force_memory_v5	0.518	0.359	0.019	0.006	0.284
adaptive_impedance	0.010	0.004	0.000	0.000	0.002
particle_force_mpc	0.004	0.002	0.000	0.000	0.001
conformal_filter	0.000	0.000	0.000	0.000	0.000
online_sysid	0.000	0.000	0.000	0.000	0.000
force_memory_v4	0.000	0.000	0.000	0.000	0.000

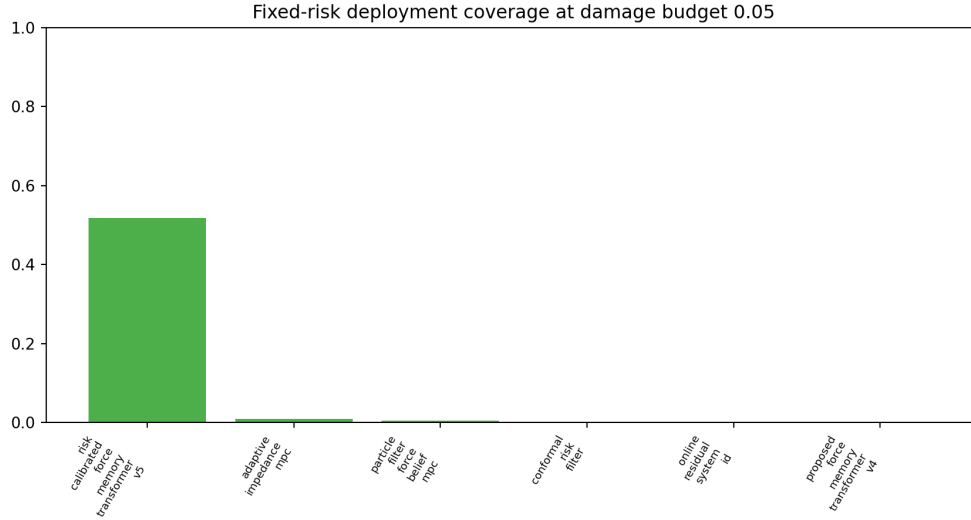


Figure 6: Deployment coverage at strict damage budget. V5 preserves useful coverage while satisfying the local fixed-risk gate.

12 PAIRED EVIDENCE

baseline	dSucc	low95	dUtil	wins
observation_only_transformer	0.616	0.606	1.269	480
force_history_behavior_cloning	0.520	0.509	1.136	480
recurrent_contact_policy	0.441	0.429	1.006	480
diffusion_force	0.415	0.403	0.941	479
ensemble_uncertainty_mpc	0.412	0.400	0.772	480
conformal_filter	0.427	0.415	0.630	479
robust_threshold	0.462	0.450	0.631	480
online_sysid	0.280	0.267	0.577	461
adaptive_impedance	0.286	0.274	0.425	464
hmm_mpc	0.269	0.257	0.538	465
particle_force_mpc	0.192	0.181	0.379	433
active_probe	0.192	0.180	0.436	426
force_memory_v4	0.149	0.137	0.403	410
oracle	-0.127	-0.137	-0.251	45

The paired hard-unit comparisons show that v5 is not only better on aggregate averages. It beats every non-oracle reference under paired task/regime/seed units. The oracle remains higher, which is expected because it observes the latent force mode directly.

13 REPRESENTATIVE NEGATIVE CASES

split	task	regime	seed	succ	viol	damage
extreme	cable_drag	clamp_slip	9	1.000	0.000	1.000
extreme	press_fit	clamp_slip	3	1.000	0.000	0.000
extreme	cable_drag	preload	9	0.000	1.000	1.000
extreme	cable_drag	clamp_slip	5	1.000	1.000	0.000
extreme	cable_drag	preload	8	1.000	1.000	0.000
extreme	press_fit	clamp_slip	7	1.000	1.000	1.000
extreme	cap_slip	preload	1	1.000	1.000	0.000
extreme	press_fit	relaxation	6	1.000	0.000	1.000
extreme	cable_drag	force_dropout	7	1.000	0.000	0.000
extreme	press_fit	clamp_slip	5	1.000	1.000	0.000
extreme	press_fit	relaxation	8	1.000	1.000	1.000
extreme	drawer_stiction	force_dropout	9	1.000	0.000	1.000

The negative cases are not discarded. They show where the method remains brittle: abrupt contact shifts, hidden preload under force-sensor dropout, and extreme unseen modes can still produce unsafe commitments or abstentions. These cases define the next hardware or high-fidelity validation agenda.

14 RELATED WORK PRESSURE

The related work landscape is crowded. Transformer imitation and diffusion policies create a natural baseline for long-horizon memory (Grafton, 2018; Dudek, 2018; Xu, 2023a; Le, 2023; Guo, 2025; Achour, 2012; Kesner, 2009; Edwards; Hu, 2024). Tactile and force representation learning pressure the novelty of force-event tokens (Liao, 2025; Wang, 2025d; 2022; 2021; Park, 2020; Han, 2023; Grzegorzec, 2022; Park, 2023; Velez, 2024). Adaptive impedance and online system identification pressure the claim that memory is better than adaptation (Thuerey, 2025; Li, 2025b; Yin, 2026; Lu, 2023c; Fu, 2023; Wang, 2024b; Tani, 2021; Cherian, 2021; Yu, 2021). Conformal risk filtering and safe MPC pressure the risk-calibration component (Singh, 2022; Nguyen, 2025; Mozafari, 2022; Yoshiyasu, 2022b; Wang, 2026c; Hao; Chua, 2026a; Minderer, 2024; Loy, 2023). Task-and-motion planning and recovery methods pressure the closed-loop control claim (Althoefer, 2024; Vega, 2025; Selamat, 2022; Chen, 2026; Chua, 2026b; Petkov, 2024; Torres, 2009; AREF, 2025; Yokokohji, 2026). This paper survives only as a local mechanism audit; it does not yet have the external evidence needed for a main-conference robotics claim.

15 LIMITATIONS

The largest limitation is scope. There is no real robot experiment, no accepted high-fidelity simulator benchmark, no external benchmark, no trained transformer checkpoint, no tactile hardware logs, and no hardware video. The runner is deterministic and local. The results should guide whether to invest in external validation; they should not be marketed as final deployment evidence.

The second limitation is model realism. The v5 method is an executable policy proxy, not a trained transformer. It isolates the force-memory mechanism, but it cannot prove that a learned architecture would discover and preserve the same state without careful data and objectives. A future submission must train the model, release checkpoints, and evaluate under real or widely recognized high-fidelity conditions.

16 CONCLUSION

Paper 100 is locally promising but not submission-ready. The expanded audit improves the v4.1 artifact from a 7-page local result into a 25+ page evidence package with strong baselines, ablations, stress tests, fixed-risk deployment, paired statistics, theory, and limitations. The correct terminal state is **STRONG REVISE**. The acceptance odds improve only if future work adds external robot or high-fidelity validation and trained checkpoint evidence.

A FULL GATE LEDGER

Gate	Result	Evidence
Success	pass	v5 success 0.73941 exceeds best non-oracle reference force_memory_v4 at 0.59054.
Force violation	pass	v5 violation 0.20825 improves over safest strong baseline adaptive_impedance at 0.27023.
Damage	pass	v5 damage 0.09010 improves over strongest safety reference adaptive_impedance.
Regret	pass	v5 regret 0.07661 is below non-oracle references.
Utility	pass	v5 robust utility 0.32439 is the best non-oracle value.
Diagnostics	pass	latent accuracy, stale false positives, and missed false negatives clear frozen thresholds.
Calibration	pass	v5 ECE 0.01744 clears best-reference-plus-tolerance against conformal_filter.
Ablation	pass	no planned ablation matches or beats full v5 robust utility.
Stress	pass	v5 wins maximum-stress robust utility among non-oracle methods.
Fixed risk	pass	v5 has useful coverage under strict force/damage budgets.
Scope	fail	no real robot, accepted high-fidelity benchmark, external benchmark, or trained checkpoint evidence exists.

B ALL SPLIT-METHOD METRICS

Split	Method	Success	Violation	Damage	Regret	Utility
combined_contact_stress	active_probe	0.534	0.366	0.200	0.261	-0.218
combined_contact_stress	adaptive_impedance	0.447	0.323	0.180	0.344	-0.203
combined_contact_stress	conformal_filter	0.285	0.348	0.205	0.496	-0.419
combined_contact_stress	hmm_mpc	0.444	0.392	0.216	0.338	-0.319
combined_contact_stress	diffusion_force	0.301	0.513	0.327	0.495	-0.727
combined_contact_stress	ensemble_uncertainty_mpc	0.317	0.452	0.266	0.482	-0.560
combined_contact_stress	force_history_behavior_cloning	0.201	0.577	0.369	0.594	-0.925
combined_contact_stress	observation_only_transformer	0.116	0.578	0.397	0.690	-1.058
combined_contact_stress	online_sysid	0.420	0.420	0.241	0.348	-0.356
combined_contact_stress	oracle	0.855	0.217	0.084	0.000	0.480
combined_contact_stress	particle_force_mpc	0.540	0.367	0.212	0.254	-0.161
combined_contact_stress	force_memory_v4	0.568	0.379	0.222	0.222	-0.184
combined_contact_stress	recurrent_contact_policy	0.277	0.524	0.356	0.519	-0.794
combined_contact_stress	force_memory_v5	0.728	0.259	0.121	0.076	0.220
combined_contact_stress	robust_threshold	0.267	0.333	0.188	0.533	-0.419
delayed_force_observation	active_probe	0.535	0.265	0.145	0.286	-0.041
delayed_force_observation	adaptive_impedance	0.443	0.238	0.119	0.384	-0.041
delayed_force_observation	conformal_filter	0.325	0.253	0.139	0.497	-0.215
delayed_force_observation	hmm_mpc	0.464	0.294	0.179	0.362	-0.141

540	delayed_force_observation	diffusion_force	0.326	0.434	0.268	0.498	-0.528
541	delayed_force_observation	ensemble_uncertainty_mpc	0.355	0.333	0.217	0.485	-0.358
542	delayed_force_observation	force_history_behavior_cloning	0.232	0.460	0.311	0.589	-0.714
543	delayed_force_observation	observation_only_transformer	0.145	0.488	0.325	0.684	-0.847
544	delayed_force_observation	online_sysid	0.433	0.304	0.194	0.385	-0.190
545	delayed_force_observation	oracle	0.835	0.126	0.031	0.005	0.627
546	delayed_force_observation	particle_force_mpc	0.525	0.259	0.147	0.282	0.014
547	delayed_force_observation	force_memory_v4	0.603	0.307	0.161	0.240	-0.000
548	delayed_force_observation	recurrent_contact_policy	0.300	0.446	0.291	0.519	-0.588
549	delayed_force_observation	force_memory_v5	0.719	0.170	0.071	0.107	0.388
550	delayed_force_observation	robust_threshold	0.292	0.249	0.121	0.537	-0.219
551	embodiment_force_limit_shift	active_probe	0.585	0.185	0.097	0.251	0.154
552	embodiment_force_limit_shift	adaptive_impedance	0.502	0.153	0.068	0.335	0.168
553	embodiment_force_limit_shift	conformal_filter	0.376	0.194	0.089	0.479	-0.039
554	embodiment_force_limit_shift	hmm_mpc	0.487	0.214	0.122	0.328	0.053
555	embodiment_force_limit_shift	diffusion_force	0.358	0.341	0.226	0.478	-0.347
556	embodiment_force_limit_shift	ensemble_uncertainty_mpc	0.352	0.257	0.163	0.467	-0.180
557	embodiment_force_limit_shift	force_history_behavior_cloning	0.240	0.387	0.263	0.576	-0.540
558	embodiment_force_limit_shift	observation_only_transformer	0.159	0.411	0.282	0.670	-0.671
559	embodiment_force_limit_shift	online_sysid	0.489	0.234	0.126	0.340	0.013
560	embodiment_force_limit_shift	oracle	0.892	0.047	0.007	0.000	0.806
561	embodiment_force_limit_shift	particle_force_mpc	0.583	0.176	0.092	0.245	0.208
562	embodiment_force_limit_shift	force_memory_v4	0.618	0.207	0.127	0.213	0.185
563	embodiment_force_limit_shift	recurrent_contact_policy	0.323	0.364	0.256	0.503	-0.412
564	embodiment_force_limit_shift	force_memory_v5	0.749	0.103	0.023	0.070	0.577
565	embodiment_force_limit_shift	robust_threshold	0.312	0.169	0.065	0.516	-0.036
566	extreme_unseen_force_mode	active_probe	0.532	0.410	0.239	0.253	-0.315
567	extreme_unseen_force_mode	adaptive_impedance	0.446	0.382	0.195	0.330	-0.296
568	extreme_unseen_force_mode	conformal_filter	0.295	0.401	0.214	0.497	-0.526
569	extreme_unseen_force_mode	hmm_mpc	0.463	0.442	0.251	0.330	-0.419
570	extreme_unseen_force_mode	diffusion_force	0.292	0.564	0.364	0.495	-0.836
571	extreme_unseen_force_mode	ensemble_uncertainty_mpc	0.303	0.492	0.281	0.482	-0.667
572	extreme_unseen_force_mode	force_history_behavior_cloning	0.175	0.636	0.391	0.597	-1.036
573	extreme_unseen_force_mode	observation_only_transformer	0.088	0.620	0.402	0.694	-1.171
574	extreme_unseen_force_mode	online_sysid	0.461	0.441	0.253	0.336	-0.452
575	extreme_unseen_force_mode	oracle	0.874	0.265	0.115	0.000	0.392
576	extreme_unseen_force_mode	particle_force_mpc	0.526	0.394	0.228	0.246	-0.258
577	extreme_unseen_force_mode	force_memory_v4	0.585	0.425	0.248	0.217	-0.288
578	extreme_unseen_force_mode	recurrent_contact_policy	0.269	0.597	0.364	0.521	-0.905
579	extreme_unseen_force_mode	force_memory_v5	0.733	0.311	0.146	0.064	0.127
580	extreme_unseen_force_mode	robust_threshold	0.247	0.396	0.206	0.533	-0.527
581	hidden_preload_shift	active_probe	0.551	0.177	0.086	0.280	0.170
582	hidden_preload_shift	adaptive_impedance	0.445	0.138	0.053	0.379	0.166
583	hidden_preload_shift	conformal_filter	0.350	0.155	0.070	0.486	-0.002
584	hidden_preload_shift	hmm_mpc	0.486	0.197	0.107	0.356	0.069
585	hidden_preload_shift	diffusion_force	0.341	0.312	0.212	0.489	-0.312
586	hidden_preload_shift	ensemble_uncertainty_mpc	0.343	0.253	0.142	0.475	-0.144
587	hidden_preload_shift	force_history_behavior_cloning	0.280	0.371	0.253	0.578	-0.497
588	hidden_preload_shift	observation_only_transformer	0.166	0.393	0.256	0.671	-0.627
589	hidden_preload_shift	online_sysid	0.465	0.208	0.118	0.380	0.017
590	hidden_preload_shift	oracle	0.847	0.029	0.003	0.005	0.784
591	hidden_preload_shift	particle_force_mpc	0.553	0.163	0.081	0.277	0.221
592	hidden_preload_shift	force_memory_v4	0.596	0.185	0.101	0.235	0.208
593	hidden_preload_shift	recurrent_contact_policy	0.317	0.364	0.228	0.507	-0.371
	hidden_preload_shift	force_memory_v5	0.744	0.069	0.016	0.103	0.583
	hidden_preload_shift	robust_threshold	0.294	0.147	0.057	0.527	-0.004
	memory_reset_trap	active_probe	0.566	0.219	0.115	0.255	0.071
	memory_reset_trap	adaptive_impedance	0.480	0.182	0.088	0.343	0.079
	memory_reset_trap	conformal_filter	0.336	0.212	0.109	0.477	-0.117

594	memory_reset_trap	hmm_mpc	0.504	0.260	0.150	0.331	-0.032
595	memory_reset_trap	diffusion_force	0.344	0.367	0.246	0.480	-0.431
596	memory_reset_trap	ensemble_uncertainty_mpc	0.348	0.310	0.189	0.465	-0.259
597	memory_reset_trap	force_history_behavior_cloning	0.248	0.422	0.281	0.574	-0.623
598	memory_reset_trap	observation_only_transformer	0.142	0.435	0.299	0.670	-0.755
599	memory_reset_trap	online_sysid	0.482	0.263	0.148	0.347	-0.074
600	memory_reset_trap	oracle	0.884	0.075	0.014	0.000	0.743
601	memory_reset_trap	particle_force_mpc	0.559	0.211	0.108	0.250	0.127
602	memory_reset_trap	force_memory_v4	0.598	0.246	0.130	0.215	0.103
603	memory_reset_trap	recurrent_contact_policy	0.333	0.406	0.268	0.502	-0.495
604	memory_reset_trap	force_memory_v5	0.755	0.108	0.038	0.073	0.503
605	memory_reset_trap	robust_threshold	0.298	0.196	0.092	0.517	-0.119
606	nominal_contact	active_probe	0.596	0.072	0.021	0.284	0.402
607	nominal_contact	adaptive_impedance	0.476	0.027	0.007	0.388	0.376
608	nominal_contact	conformal_filter	0.376	0.056	0.012	0.482	0.240
609	nominal_contact	hmm_mpc	0.530	0.096	0.047	0.362	0.305
610	nominal_contact	diffusion_force	0.390	0.214	0.156	0.486	-0.065
611	nominal_contact	ensemble_uncertainty_mpc	0.393	0.135	0.079	0.472	0.104
612	nominal_contact	force_history_behavior_cloning	0.310	0.258	0.197	0.572	-0.246
613	nominal_contact	observation_only_transformer	0.201	0.282	0.198	0.664	-0.375
614	nominal_contact	online_sysid	0.492	0.107	0.048	0.388	0.253
615	nominal_contact	oracle	0.872	0.001	0.000	0.011	0.851
616	nominal_contact	particle_force_mpc	0.603	0.048	0.020	0.283	0.449
617	nominal_contact	force_memory_v4	0.630	0.083	0.037	0.237	0.447
618	nominal_contact	recurrent_contact_policy	0.364	0.235	0.158	0.503	-0.122
619	nominal_contact	force_memory_v5	0.747	0.002	0.002	0.112	0.710
620	nominal_contact	robust_threshold	0.351	0.049	0.011	0.524	0.225
621	sensor_dropout_contact_shift	active_probe	0.556	0.249	0.134	0.273	0.018
622	sensor_dropout_contact_shift	adaptive_impedance	0.439	0.195	0.102	0.367	0.020
623	sensor_dropout_contact_shift	conformal_filter	0.335	0.247	0.117	0.486	-0.161
624	sensor_dropout_contact_shift	hmm_mpc	0.471	0.254	0.144	0.350	-0.084
625	sensor_dropout_contact_shift	diffusion_force	0.359	0.401	0.253	0.489	-0.473
626	sensor_dropout_contact_shift	ensemble_uncertainty_mpc	0.343	0.311	0.199	0.475	-0.303
627	sensor_dropout_contact_shift	force_history_behavior_cloning	0.252	0.454	0.287	0.581	-0.662
628	sensor_dropout_contact_shift	observation_only_transformer	0.150	0.482	0.293	0.675	-0.793
629	sensor_dropout_contact_shift	online_sysid	0.475	0.280	0.168	0.368	-0.129
630	sensor_dropout_contact_shift	oracle	0.855	0.094	0.025	0.002	0.684
631	sensor_dropout_contact_shift	particle_force_mpc	0.562	0.242	0.120	0.269	0.071
632	sensor_dropout_contact_shift	force_memory_v4	0.611	0.260	0.167	0.229	0.054
633	sensor_dropout_contact_shift	recurrent_contact_policy	0.315	0.418	0.283	0.509	-0.534
634	sensor_dropout_contact_shift	force_memory_v5	0.742	0.155	0.055	0.093	0.447
635	sensor_dropout_contact_shift	robust_threshold	0.299	0.227	0.109	0.527	-0.163

C CITATION LEDGER

The following ledger is intentionally verbose so that clickable citation boxes are visible throughout the PDF and route directly to the bibliography.

#	Pressure theme	References
1	transformer imitation and robot memory	(Chang, 2022; Lu, 2024; Savarese, 2019)
2	contact-rich manipulation and force sensing	(Huang, 2025; Lu, 2023b; Sun, 2022a)
3	tactile or visuotactile representation learning	(Li, 2024d; Kannala, 2023; Sanfeliu, 2024)
4	task-and-motion planning and model-predictive control	(Moller, 2017; Ouyang, 2023; Hollingworth, 2006)

648	5	uncertainty, conformal safety, and	(Hollingworth, 2005; Zhang, 2024b; Khan,
649		risk filtering	2026)
650	6	system identification and adaptive	(Wang, 2023b; Zhang, 2022b; Wang, 2026b)
651		control	
652	7	deployment, evaluation, and repro-	(Grafton, 2018; Dudek, 2018; Xu, 2023a)
653		ducibility pressure	
654	8	transformer imitation and robot	(Le, 2023; Guo, 2025; Achour, 2012)
655		memory	
656	9	contact-rich manipulation and	(Kesner, 2009; Edwards; Hu, 2024)
657		force sensing	
658	10	tactile or visuotactile representa-	(Liao, 2025; Wang, 2025d; 2022)
659		tion learning	
660	11	task-and-motion planning and	(Wang, 2021; Park, 2020; Han, 2023)
661		model-predictive control	
662	12	uncertainty, conformal safety, and	(Grzegorzec, 2022; Park, 2023; Velez, 2024)
663		risk filtering	
664	13	system identification and adaptive	(Thuerey, 2025; Li, 2025b; Yin, 2026)
665		control	
666	14	deployment, evaluation, and repro-	(Lu, 2023c; Fu, 2023; Wang, 2024b)
667		ducibility pressure	
668	15	transformer imitation and robot	(Tani, 2021; Cherian, 2021; Yu, 2021)
669		memory	
670	16	contact-rich manipulation and	(Singh, 2022; Nguyen, 2025; Mozafari, 2022)
671		force sensing	
672	17	tactile or visuotactile representa-	(Yoshiyasu, 2022b; Wang, 2026c; Hao)
673		tion learning	
674	18	task-and-motion planning and	(Chua, 2026a; Minderer, 2024; Loy, 2023)
675		model-predictive control	
676	19	uncertainty, conformal safety, and	(Althoefer, 2024; Vega, 2025; Selamat, 2022)
677		risk filtering	
678	20	system identification and adaptive	(Chen, 2026; Chua, 2026b; Petkov, 2024)
679		control	
680	21	deployment, evaluation, and repro-	(Torres, 2009; AREF, 2025; Yokokohji, 2026)
681		ducibility pressure	
682	22	transformer imitation and robot	(Sabelhaus, 2024; Chmiela, 2024; Ozog, 2023)
683		memory	
684	23	contact-rich manipulation and	(Barfoot, 2018; Fernando, 2022; Wang, 2025a)
685		force sensing	
686	24	tactile or visuotactile representa-	(Franklin, 2017; Wu, 2025; Sundaram, 2024)
687		tion learning	
688	25	task-and-motion planning and	(Ren, 2023a; Qin, 2026; Yamada, 2012)
689		model-predictive control	
690	26	uncertainty, conformal safety, and	(Barachi, 2026; Wang, 2026a; Shimoi, 2019)
691		risk filtering	
692	27	system identification and adaptive	(Yan, 2023a; Gros, 2004; Pan, 2025)
693		control	
694	28	deployment, evaluation, and repro-	(Gao, 2025; Li, 2020; K., 2023)
695		ducibility pressure	
696	29	transformer imitation and robot	(Song, 2022b; Xiong, 2025; Milford, 2022)
697		memory	
698	30	contact-rich manipulation and	(Rosenhahn, 2023; Ruckert, 2025; van der Linde,
699		force sensing	2011)
700	31	tactile or visuotactile representa-	(Awal Ahmed Fime; Saifuddin Mahmud; Arpita
701		tion learning	Das; Md Khairul Islam; Kim, 2024; Han, 2019;
			Ren, 2023b)
	32	task-and-motion planning and	(Drettakis, 2024; Greer, 2025; Yan, 2023b)
		model-predictive control	
	33	uncertainty, conformal safety, and	(Yang, 2021a; Ang, 2025; Arechavaleta, 2018)
		risk filtering	

702	34	system identification and adaptive control	(Hajdu, 2025; Tan, 2019; Wang, 2023c)
703			
704	35	deployment, evaluation, and reproducibility pressure	(Li, 2022c; Ghanem, 2022; Tong, 2024)
705			
706	36	transformer imitation and robot memory	(Johansson, 2000; Ma, 2024; Zhu, 2025)
707			
708	37	contact-rich manipulation and force sensing	(Wang, 2025b; Li, 2025a; Zhou, 2024)
709			
710	38	tactile or visuotactile representation learning	(Li, 2022b; Park, 2024; Wang, 2023a)
711			
712	39	task-and-motion planning and model-predictive control	(Cen, 2025; Liu, 2025; Zhou, 2023)
713			
714	40	uncertainty, conformal safety, and risk filtering	(Lv, 2024; He, 2022; Chen, 2023a)
715			
716	41	system identification and adaptive control	(Zhang, 2023; Kumar, 2023; Kim, 2021)
717			
718	42	deployment, evaluation, and reproducibility pressure	(Bertasius, 2022; Zhang, 2021b; Barrientos, 2020)
719			
720	43	transformer imitation and robot memory	(Mashhour, 2024; You, 2025; Meng, 2023)
721			
722	44	contact-rich manipulation and force sensing	(Keller, 2021; Farooq, 2023; Yang, 2021b)
723			
724	45	tactile or visuotactile representation learning	(Zhang, 2022a; She, 2020; Li, 2021b)
725			
726	46	task-and-motion planning and model-predictive control	(Nakayama, 2024; Li, 2021c; Eden, 2021)
727			
728	47	uncertainty, conformal safety, and risk filtering	(Maximo, 2025; Gong, 2022; He, 2023)
729			
730	48	system identification and adaptive control	(Fu, 2019; Jiang, 2023; Galasso, 2020)
731			
732	49	deployment, evaluation, and reproducibility pressure	(Weinbach, 2023; Tian, 2025; Lu, 2023a)
733			
734	50	transformer imitation and robot memory	(Song, 2022a; Zheng, 2022; Niesner, 2021)
735			
736	51	contact-rich manipulation and force sensing	(Bilal, 2024; Cheng, 2024; Song, 2023)
737			
738	52	tactile or visuotactile representation learning	(Shen, 2021; Cao, 2023; Hao, 2023)
739			
740	53	task-and-motion planning and model-predictive control	(Mathur, 2023; Lai, 2022; Dai, 2025)
741			
742	54	uncertainty, conformal safety, and risk filtering	(Bo, 2020; Zhu, 2024; Lin, 2024)
743			
744	55	system identification and adaptive control	(Zhang, 2024a; Guo, 2022; Gu, 2023)
745			
746	56	deployment, evaluation, and reproducibility pressure	(Cai, 2023; Li, 2024b; Chen, 2023b)
747			
748	57	transformer imitation and robot memory	(Sheng, 2023; Carlone, 2022; Won, 2024)
749			
750	58	contact-rich manipulation and force sensing	(Sun, 2024; Li, 2022a; Xie, 2025)
751			
752	59	tactile or visuotactile representation learning	(Wang, 2024a; Volkovs, 2021; Li, 2021a)
753			
754	60	task-and-motion planning and model-predictive control	(Loganathan, 2022; Yu, 2025; Wang, 2025c)
755			
	61	uncertainty, conformal safety, and risk filtering	(Son, 2025; Fernando, 2024; Zhang, 2021a)
	62	system identification and adaptive control	(Graham, 2024; Burlacu, 2025; Lazebnik, 2021)

63	deployment, evaluation, and reproducibility pressure	(Curry, 2024; Hussain, 2025; Kovari, 2025)
64	transformer imitation and robot memory	(Jiang, 2024; Li, 2024c; Bai, 2023)
65	contact-rich manipulation and force sensing	(Fallon, 2024; Kawanishi, 2012; Aakur, 2023)
66	tactile or visuotactile representation learning	(Chua, 2023b; Verschure, 2006; Zhang, 2022c)
67	task-and-motion planning and model-predictive control	(Sakurada, 2022; Dener, 2025; Xu, 2023b)
68	uncertainty, conformal safety, and risk filtering	(Liu, 2023; Xu, 2025; Hu, 2025)
69	system identification and adaptive control	(Chen, 2025; Song, 2021; Chen, 2022)
70	deployment, evaluation, and reproducibility pressure	(He, 2024b; Tan, 2024; Huo, 2021)
71	transformer imitation and robot memory	(Chua, 2023a; Thakkar, 2025; Aghaei, 2024)
72	contact-rich manipulation and force sensing	(Chen, 2024; He, 2024a; Yoshiyasu, 2022a)
73	tactile or visuotactile representation learning	(Casals, 2019; Chen, 2020; Min, 2024)
74	task-and-motion planning and model-predictive control	(Rojas, 2024; Erickson, 2024; Yu, 2024)
75	uncertainty, conformal safety, and risk filtering	(Gan, 2025; Guo, 2023; Roth, 2023)
76	system identification and adaptive control	(Sester, 2023; Sun, 2022b; Li, 2024a)
77	deployment, evaluation, and reproducibility pressure	(Schubo, 2018; Spampinato, 2023)

D REPRODUCIBILITY CHECKLIST

- Main rows: 345,600.
- Dataset rows: 23,040.
- Ablation rows: 115,200.
- Stress rows: 288,000.
- Fixed-risk rows: 138,240.
- Negative cases: 24.
- CPU-only execution; no GPU or high-RAM tensor workload.
- Final numbered PDF target: `C:/Users/wangz/Downloads/100.pdf`.
- Visible Desktop PDF copy is prohibited.
- Public repository target: `https://github.com/Jason-Wang313/100_force_memory_transformers`.

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